

RGB Colour GENERATOR

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Presented here is a red, green and blue (RGB) colour generator using AT89C2051 microcontroller (MCU). This is a

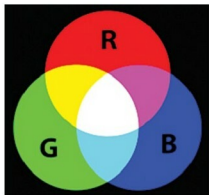


Fig. 1: Concept of primary colour mixing

simple and low-cost circuit for a multiple colour generator that can be built easily with a few additional components. The main objective of the project is to generate multiple colours from RGB primary colours as used in television displays, computer monitors and other colour displays for commercial applications like LED projectors. The concept of primary colour mixing is shown in Fig. 1.

Additive mixing of red and green lights produces shades of yellow, orange or brown. Mixing green and blue produces shades of cyan, whereas mixing red and blue produces shades of purple including magenta. Mixing nominally equal

proportions of the three primaries results in shades of grey or white; colour space that is generated is called an RGB colour space.

Circuit and working

The circuit diagram for the multiple colour generator is shown in Fig. 2. The circuit requires a 9V battery, 7805 voltage regulator (IC1), Atmel AT89C2051 MCU (IC2) and a few other components. The 20-pin MCU performs the operation of multiple colour generation from RGB primary colours on pressing switches S1 through S4. In this project, we have used the concept of pulse width modulation (PWM) to change the colours of the RGB LED.

Initially, the circuit is powered by a +9V supply, which is connected to IC1 to maintain a constant +5V at its output. This constant output is fed to the MCU circuit. Port3 of

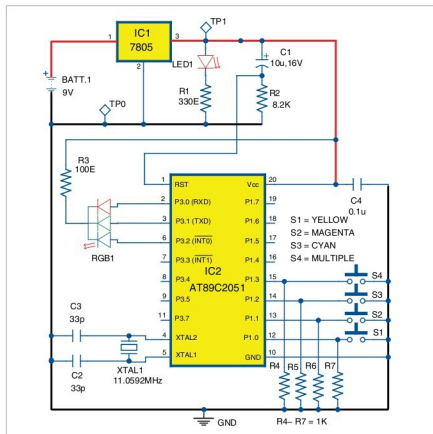


Fig. 2: Circuit of the RGB colour generator

Test Points

Test point	Details
TP0	0V GND
TP1	+5V

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V voltage regulator
IC2 - AT89C2051 MCU with a

- | | |
|------|---------------|
| LED1 | - 5mm LED |
| RGB1 | - 5mm RGB LED |

Resistors (all 1/4

- | | |
|-------|----------------|
| R1 | - 330-ohm |
| R2 | - 8.2-kilo-ohm |
| R3 | - 100-ohm |
| R4-R7 | - 1-kilo-ohm |

Capacitors:

- C1 - 10 μ F, 16V electrolytic
C2, C3 - 33pF ceramic
C4 - 0.1 μ F ceramic

Miscellaneous:

- | | |
|-------|--|
| S1-S4 | - Tactile switch |
| XTAL1 | - 11.0592MHz crystal oscillator |
| CON1 | - 2-pin connector terminal
- 9V battery |